

Linear Inequalities

1. [June 2023]

In the xy -plane, which of the following does NOT contain any points (x, y) that are solutions to $7x + 4y > 12$?

- A) The region where $x > 0$ and $y > 0$
- B) The region where $x < 0$ and $y > 0$
- C) The region where $x < 0$ and $y < 0$
- D) The region where $x > 0$ and $y < 0$

2. [August US 2023]

$$y \leq \frac{2}{3}x + 5$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A)

x	y
-1.5	6
1.5	5
3	-4

B)

x	y
-1.5	5
1.5	7
3	8

C)

x	y
-1.5	6
1.5	5
3	6

D)

x	y
-1.5	3
1.5	5
3	6

3. [June 2023]

In December 2017, the lowest temperature recorded in a certain city was 40 degrees Fahrenheit (F°) and the highest temperature recorded was $90^\circ F$. Which inequality is true for all values of t , where t represents any temperature in F° , recorded in the city in December?

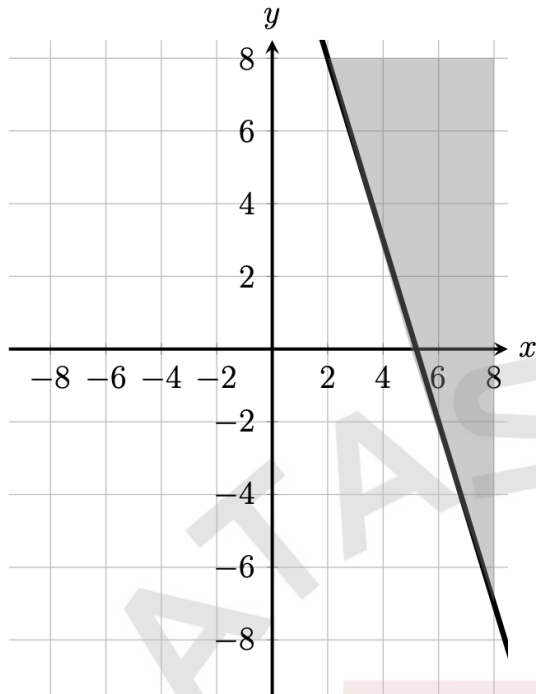
- A) $40 \leq t \leq 90$
- B) $t \leq 40$
- C) $t \leq 50$
- D) $t \geq 90$

4. [December 2023]

A model estimates that a gray whale travels 73 to 78 miles each day during its migration. Based on this model, which inequality represents the estimated total number of miles, x , a gray whale could travel in 31 days of its migration?

- A) $(73)(31) \leq x \leq (78)(31)$
- B) $73 + 31 \leq x \leq 78 + 31$
- C) $73 \leq 31x \leq 78$
- D) $73 \leq 31 + x \leq 78$

5. [December 2023]



The shaded region shown represents solutions to inequality. Which ordered pair (x, y) is a solution to this inequality?

- A) $(-7, 0)$
- B) $(0, -7)$
- C) $(0, 7)$
- D) $(7, 0)$

6. [March 2024]

The minimum value of x is 11 less than 8 times another number n . Which inequality shows the possible values of x ?

- A) $x \leq 8n - 11$
- B) $x \geq 8n - 11$
- C) $x \leq 11 - 8n$
- D) $x \geq 11 - 8n$

7. [November 2023]

$$y \leq x + 9$$

$$y \geq -2x - 9$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

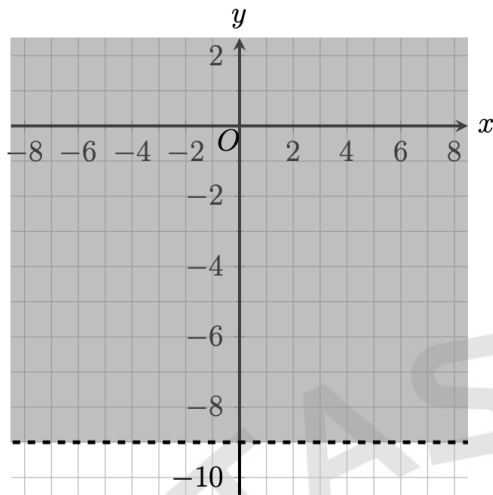
- A) $(0, -10)$
- B) $(0, 10)$
- C) $(-7, 0)$
- D) $(7, 0)$

8. [May 2024]

An arborist measured and recorded the heights of red maple trees in a certain area. Of the red maple tree heights recorded in this area, the minimum was 42 feet and the maximum was 57 feet. Which inequality is true for all possible values of t , where t represents the recorded height, in feet, of a red maple tree in this area?

- A) $t \leq 42$
- B) $t \geq 57$
- C) $t \geq 99$
- D) $42 \leq t \leq 57$

9. [March 2024]



The shaded region shown represents the solutions to the inequality $-18y < c$, where c is a constant. What is the value of c ?

- A) 162
- B) 9
- C) -9
- D) -162

10. [March 2024]

$$y \leq x + 2$$

$$y \geq -3x - 4$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

- A) $(0, -8)$
- B) $(0, 8)$
- C) $(-8, 0)$
- D) $(8, 0)$

11. [March 2024]

A certain truck can tow a trailer if the combined weight of the trailer and the packages it contains is no more than 4600 pounds. What is the maximum number of objects this truck can tow in a trailer with a weight of 600 pounds if each package weighs 120 pounds?

- A) 33
- B) 34
- C) 38
- D) 39

12. [March 2024]

A certain truck can tow a trailer if the combined weight of the trailer and the packages it contains is no more than 4,200 pounds. What is the maximum number of packages the truck can tow in a trailer with a weight of 600 pounds if each package weighs 110 pounds?

- A) 32
- B) 33
- C) 38
- D) 39

13. [March 2024]

$$y \leq x + 2$$

$$y \geq -4x - 6$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

- A) $(0, -12)$
- B) $(0, 12)$
- C) $(-12, 0)$
- D) $(12, 0)$

14. [May US 2024]

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of 18 and the fourth odd integer is at most 22 less than the sum of the first and third odd integers. Which inequality represents this situation?

- A) $18(x + 6) \leq x + (x + 4) - 22$
- B) $18(x + 6) \geq 22 - (x + (x + 4))$
- C) $18(x + 4) \leq x + (x + 3) - 22$
- D) $18(x + 4) \geq 22 - (x + (x + 3))$

15. [June 2024]

During a portion of a flight, a small airplane's cruising speed varied between 180 miles per hour and 190 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

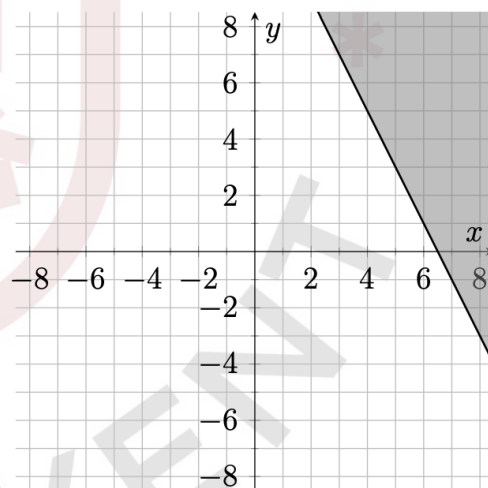
- A) $s \leq 10$
- B) $s \leq 180$
- C) $s \leq 190$
- D) $180 \leq s \leq 190$

16. [June 2024]

During a portion of a flight, a small airplane's cruising speed varied between 125 miles per hour and 135 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

- A) $s \leq 10$
- B) $s \leq 125$
- C) $s \leq 135$
- D) $125 \leq s \leq 135$

17. [June 2024]



The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

- | | |
|--------------|--------------|
| A) $(-8, 0)$ | C) $(0, -8)$ |
| B) $(0, 8)$ | D) $(8, 0)$ |

18. [August 2024]

A newspaper stand sells magazines for \$5 each and newspapers for \$2 each. The stand needs to sell m magazines and n newspapers for a combined total value of at least \$150 each day to cover operation costs. Which inequality represents this situation?

- A) $5m + 2n \leq 150$
- B) $5m + 2n \geq 150$
- C) $7mn \leq 150$
- D) $7mn \geq 150$

19. [August 2024]

$y < -6x - 22$
 $y > -4x - 7$

For what of the following tables are all the values of x and their corresponding values of y solutions to the given system of inequalities?

- A)

x	-9	-10	-12
y	27	51	-4
- B)

x	-9	-10	-12
y	32	36	-4
- C)

x	-9	-10	-12
y	28	36	48
- D)

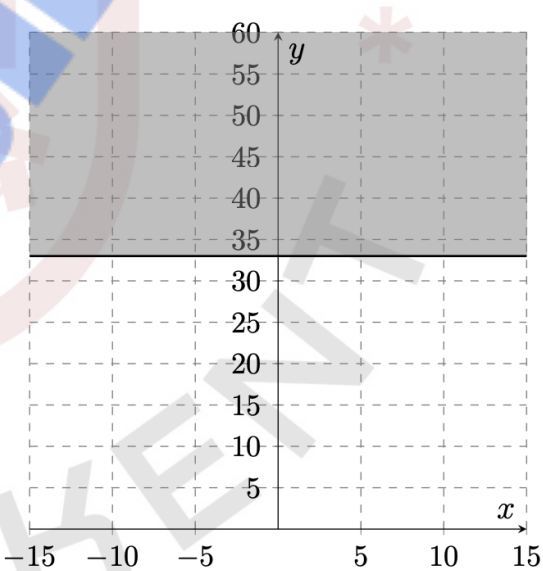
x	-9	-10	-12
y	30	37	48

20. [August 2024]

The water depth in a certain river is 78.7 feet (ft). During a 30-day period, the water depth in this river is predicted to increase by a minimum of 1 ft and a maximum of 3 ft. What is the predicted maximum water depth, in feet, during this time?

- A) 75.7
- B) 77.7
- C) 79.7
- D) 81.7

21. [August US 2024]



The shaded region shown in the graph represents all the solutions to which inequality?

- A) $x < 33$
- B) $x \geq 33$
- C) $y \leq 33$
- D) $y \geq 33$

22. [August 2024]

A model estimates that a gray whale travels 72 to 77 miles each day during its migration. Based on this model, which inequality represents the estimated total number of miles, x , a gray whale could travel in 14 days of its migration?

- A) $(72)(14) \leq x \leq (77)(14)$
- B) $72 + 14 \leq x \leq 77 + 14$
- C) $72 \leq 14x \leq 77$
- D) $72 \leq 14 + x \leq 77$

23. [October US 2024]

A plant nursery classifies a perennial species as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represent the possible heights h , in centimeters, for a specific perennial species when fully grown. Which inequality represents the possible heights h , in centimeters, of a tall perennial flower species?

- A) $107 < h < 158$
- B) $76 < h < 100$
- C) $41 < h < 95$
- D) $16 < h < 91$

24. [October US 2024]

A shopper goes to a pet store to buy dog food and treats. The shopper has \$20. Each can of food costs \$1.87, and each bag of treats costs \$2.37. Which inequality represents the relationship between the number of cans of food, c , and the number of bags of treats, t , the shopper can buy?

- A) $1.87c + 2.37t \leq 20$
- B) $1.87c + 2.37t \geq 20$
- C) $2.37c + 1.87t \leq 20$
- D) $2.37c + 1.87t \geq 20$

25. [October 2024]

A geologist estimates that the volume of a boulder is greater than 12.7 cubic feet but less than 15.7 cubic feet. The geologist also estimates that the boulder weighs 160 pounds per cubic foot of volume. Which inequality represents this situation, where x represents the estimated total weight, in pounds, of the boulder?

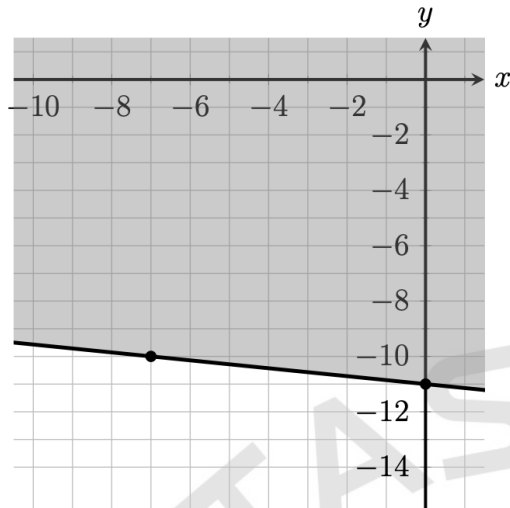
- A) $160 - 15.7 < x < 160 - 12.7$
- B) $160 + 12.7 < x < 160 + 15.7$
- C) $160(12.7) < x < 160(15.7)$
- D) $\frac{160}{15.7} < x < \frac{160}{12.7}$

26. [October US 2024]

For a particular tree species, it is recommended to plant a maximum of 102 trees per acre. Based on this recommendation, what is the maximum number of trees of this species that can be planted on 3 acres?

- A) 408
- B) 306
- C) 105
- D) 99

27. [October US 2024]



The shaded region shown represents solution to $rx + ty \geq -77$, where r and t are constants. What is the value of $r + t$?

- A) 8
- B) 6
- C) -6
- D) -7

28. [October US 2024]

A botanist estimates the number of oak trees in a forest to be at least 2600 and no more than 3200. Which inequality best represents the number oak trees, x , in the forest?

- A) $x \leq 600$
- B) $x \leq 2600$
- C) $2600 \leq x \leq 3200$
- D) $x \leq 2600$ or $x \geq 3200$

29. [October US 2024]

If $x \geq -6$ represent all solutions to the inequality $ax - 27 \leq 15$, where a is a constant, what is the greatest possible value of a ?

30. [November 2024]

Cristina was elected to the city council. Each city council person is required to attend at least 21 of the 27 scheduled meetings in a year to keep their position. Which inequality represents the number of city council meetings, x , Cristina must attend in a year to keep her position on the city council?

- A) $x \leq 21$
- B) $x \geq 48$
- C) $21 \leq x \leq 27$
- D) $27 \leq x \leq 48$

31. [November 2024]

To cut a lawn, Antwan charges a fee of \$10.00 for his equipment and \$8.50 per hour spent cutting lawn. Taylor charges a fee of \$7.00 for his equipment and \$9.50 per hour spent cutting a lawn. If x represents the number of hours spent cutting a lawn, what are the value of x for which Taylor's total charge is greater than Antwan's total charge?

- A) $2 \leq x \leq 3$
- B) $3 \leq x \leq 4$
- C) $x < 2$
- D) $x > 3$

32. [November US 2024]

An employee earns \$30 per hour for the first 40 hours worked in a week. For each hour worked over 40 hours, the employee earns \$45 per hour. What is the least number of whole hours that the employee must work in a week to earn at least \$1,610?

- A) 50 C) 10
B) 49 D) 9

33. [November US 2024]

$$y > 5x - 6$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A)

x	y
3	3
5	13
8	28

B)

x	y
3	9
5	19
8	34

C)

x	y
3	11
5	21
8	39

D)

x	y
3	-9
5	-19
8	-34

34. [December US 2024]

$$y > 17$$

$$4x + y < 21$$

The point $(x, 54)$ is a solution to the given system of inequalities in the xy -plane. Which of the following could be the value of x ?

- A) 9 C) -5
B) 5 D) -9

35. [December US 2024]

A soccer team's goal is to earn at least \$1,320 by selling coupon books. The team earns \$12 from selling each coupon book. Which of the following inequalities describes the possible values for the number of coupon books, n , the team can sell to meet the goal?

- A) $n + 12 \leq 1320$
B) $n + 12 \geq 1320$
C) $12n \leq 1320$
D) $12n \geq 1320$

36. [December 2024]

When several resistors are connected in series in a circuit, the total resistance of the circuit is the sum of the resistances of these resistors. A certain circuit consists of 8 resistors connected in series, where the resistance of each resistor is positive. This circuit has a total resistance of 130 ohms. The total resistance of 3 resistors in this circuit is 90 ohms. Which inequality best represents all possible values of the resistance x , in ohms, of one of the other 5 resistors in this circuit?

- A) $0 < x < 8$
B) $0 < x < 40$
C) $40 < x < 90$
D) $40 < x < 130$

37. [December 2024]

Sabrina has a goal to save at least 240 USD. Sabrina has already saved 180 USD. If x represents the additional amount of money, in dollars, Sabrina needs to save to meet her goal, which inequality represents this situation?

- A) $\frac{x}{180} \geq 240$
- B) $180x \geq 240$
- C) $180 + x \geq 240$
- D) $180 - x \geq 240$

38. [December 2024]

$$y < 42 - 7x$$

$$\frac{y}{7} > 10$$

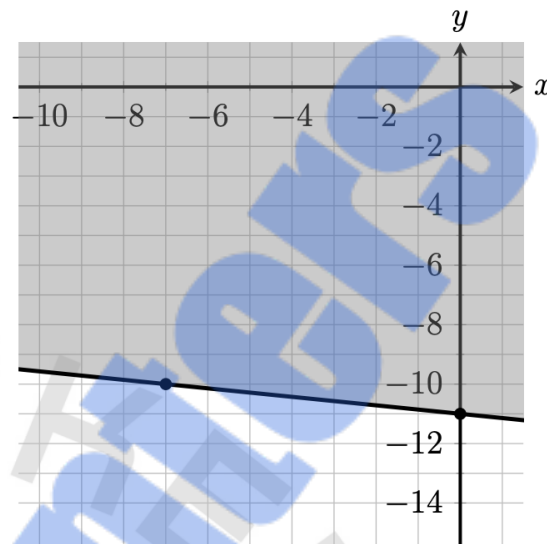
Which inequality represents the x values for all solutions (x, y) that satisfy the given system of inequalities in the xy -plane?

- A) $x > 6$
- B) $x < 6$
- C) $x > -4$
- D) $x < -4$

39. [March 2025]

A school is ordering tablet computers for its students. The tablets cost \$164 each, and 6% tax is added to the total cost of the order. If the school can spend no more than 44,800 dollars on the tablets, including tax, what is the maximum number of tablets that the school can order?

40. [March 2025]



The shaded region shown represents the solutions to $rx + ty \geq -77$, where r and t are constants. What is the value of $r + t$?

- A) 8
- B) 6
- C) -6
- D) -7

41. [March US 2025]

A shopper is purchasing oranges that cost \$0.67 each. If the shopper has a coupon for \$2 off the entire purchase and there is no sales tax, which inequality represents the number of oranges, x , that the shopper can purchase for no more than \$12?

- A) $0.67x + 2 \leq 12$
- B) $0.67x + 2 \geq 12$
- C) $0.67x - 2 \leq 12$
- D) $0.67x - 2 \geq 12$

42. [March US 2025]

$$y \leq 5x - 21$$

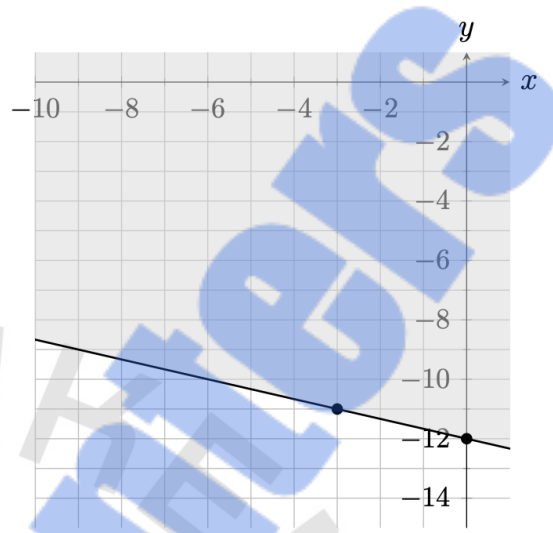
Which point (x, y) is a solution to the given inequality in the xy -plane?

- | | |
|----------------|---------------|
| A) $(-5, -21)$ | C) $(5, -21)$ |
| B) $(0, 21)$ | D) $(6, 21)$ |

43. [March US 2025]

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of 28 and the third odd integer in the set is at most the value of 50 less than the sum of the first and fourth odd integers in the set. What is the greatest possible value of x ?

45. [March 2025]



The shaded region shown represents the solutions to $rx + ty \geq -36$, where r and t are constants. What is the value of $r + t$?

- | | |
|------|-------|
| A) 4 | C) -2 |
| B) 2 | D) -4 |

44. [March 2025]

A school is ordering tablet computers for its students. The tablets cost \$170 each, and 7% tax is added to the total cost of the order. If the school can spend no more than \$53,400 on the tablets, including tax, what is the maximum number of tablets that the school can order?

46. [May 2025]

$$y \leq x + 3$$

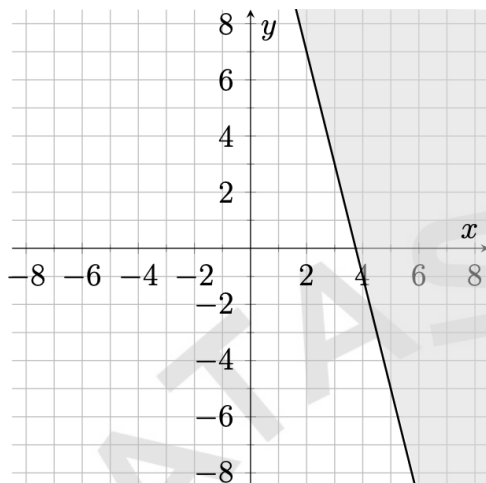
$$y \geq -3x - 7$$

Which point (x, y) is a solution to the given system of inequalities in the xy -plane?

- | | |
|---------------|---------------|
| A) $(0, -14)$ | C) $(-14, 0)$ |
| B) $(0, 14)$ | D) $(14, 0)$ |

47. [March 2025]

The shaded region shown represents solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?



- A) $(-5, 0)$
 B) $(0, 5)$
 C) $(0, -5)$
 D) $(5, 0)$

48. [May 2025]

A landscaper will plant two types of shrubs, hydrangeas and winter hazels, in a park. There will be no more than 244 total shrubs planted, and the number of hydrangeas planted will be at most three times the number of winter hazels planted. Which of the following systems of inequalities best represents this situation, where h is the number of hydrangeas that will be planted and w is the number of winter hazels that will be planted?

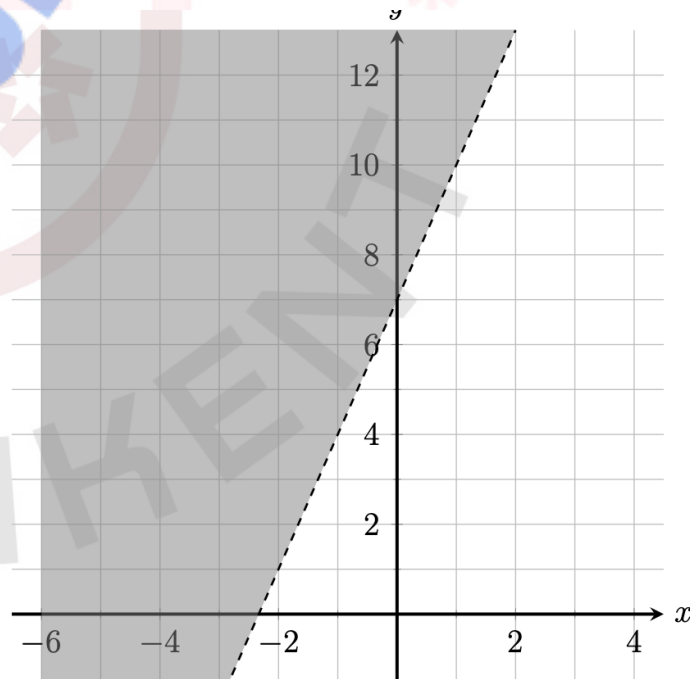
- A) $h + w \leq 244$
 $h \leq 3w$
 B) $h + w \leq 244$
 $3h \geq w$
 C) $h + w \geq 244$
 $h \leq 3w$
 D) $h + w \geq 244$
 $3h \geq w$

49. [May 2025]

For a particular tree species, it is recommended to plant a maximum of 92 trees per acre. Based on this recommendation, what is the maximum number of trees of this species that can be planted on 2 acres?

- A) 276
 B) 184
 C) 94
 D) 90

50. [May 2025]



The shaded region shown represents the solutions to which quality?

- A) $y > 3x + 7$
 B) $y > 3x - 7$
 C) $y < 3x + 7$
 D) $y < 3x - 7$

51. [May 2025]

$$y > 5x - 4$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

A)

x	y
4	11
6	21
9	36

B)

x	y
4	16
6	26
9	41

C)

x	y
4	21
6	31
9	46

D)

x	y
4	21
6	21
9	41

52. [June 2025]

A plant nursery classifies a perennial flower species as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights h , in centimeters, for a specific perennial flower species when fully grown. Which inequality represents the possible heights h , in centimeters, for a tall perennial flower species?

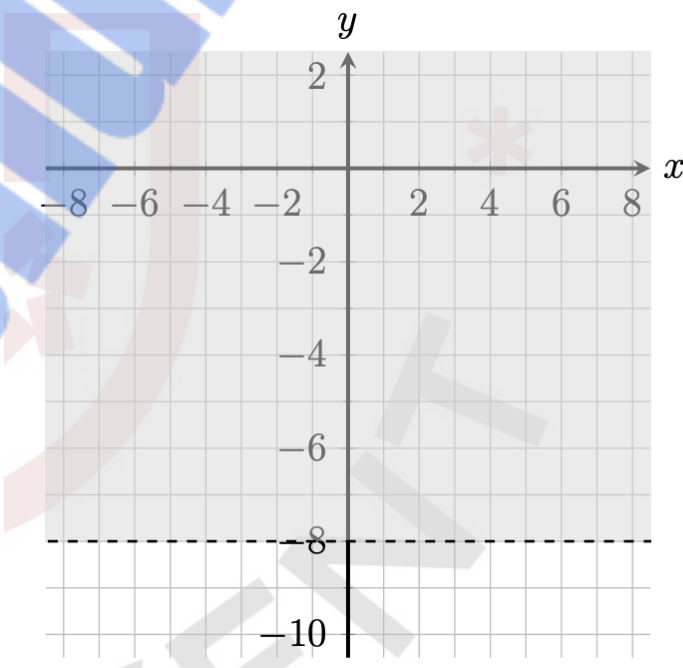
- A) $18 < h < 95$
- B) $41 < h < 89$
- C) $84 < h < 100$
- D) $110 < h < 158$

53. [June US 2025]

For a certain classification, the positive mass m , in grams, of an object must be less than 510 grams. Which inequality represents this situation?

- A) $m > 510$
- B) $m < 0$
- C) $-510 < m < 0$
- D) $0 < m < 510$

54.



The shaded region shown represents the solutions to the inequality $ry < 128$, where r is a constant. What is the value of r ?

- A) 16
- B) 8
- C) -8
- D) -16

55. [June US 2025]

A car travels at a speed of at least 25 miles per hour but no more than 50 miles per hour for a certain part of a trip. Which inequality represents this situation, where x is the speed of the car, in miles per hour, on this part of the trip?

- A) $x \geq 25$ C) $25 \leq x \leq 50$
B) $x \geq 50$ D) $x \leq 75$

56. [October US 2024]

In a set of four consecutive odd integers, where the integers are ordered from least to greatest, the first integer is represented by x . The product of 28 and the third odd integer in the set is at most the value of 50 less than the sum of the first and fourth odd integers in the set. What is the greatest possible value of x ?